

can adjust Physique parameters and then scrub the timeline to instantly see the results.

8.6.2 Applying Physique Vertices

The Vertices sub-object determines how each vertex of the mesh will be assigned and how it will deform. Vertices can be assigned to one of three categories: Deformable, Rigid, or Root. Deformable vertices accept deformation from attached nodes and will flex and bend. Rigid vertices move with the attached joint but do not deform. Root vertices do not move with the skeleton.

Deformable – Deformable vertices are indicated in red.

Rigid – Rigid vertices are indicated in green.

Fixed – Fixed vertices are indicated in blue.

Blending Between Links – Determines how the joints affect the vertices. The default is N Links, which allows any link to affect a vertex. This can be changed to limit the how many joints affect a vertex.

Select - Selects vertices for modification.

Red vertices deform, such as the vertices in the arm. Green vertices move with the joint but remain rigid, such as the head and Blue vertices do not move with the mesh. Looking for blue vertices is a good way to chase down unassigned vertices.

8.6.3 Envelopes

Envelopes are the Physique modifier's primary tool for controlling skin deformation. They are accessed by using the Envelopes sub-object. Envelopes work very similarly to the envelopes found in the Skin modifier. Each joint produces a pill-shaped envelope that surrounds the mesh and includes those vertices close to the joint. The size and shape of the envelope can be adjusted. Envelopes can also be set as deformable or rigid. When envelopes overlap, the weighting is balanced between the envelopes. Envelopes have three types of controls: Link, Cross Section, and Control Point. Link is the entire envelope, Cross Section is the radial cross section of the envelope, and Control Points are the points that control the shape of the cross section. Envelopes surround a joint and include the vertices close to the joint.

8.6.4 Bulges, Creases, and Other Deformations

In addition to envelopes, Physique has several other methods to deform the mesh based on the movement of the skeleton. Bulges allow the skin to bulge as the joints move. Tendons link one joint to another and spread